

Kelompok 2 (Analisis Real)

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Teorema:

Jika $a, b, c \in R$, maka:

$$(1) a \neq 0 \Rightarrow \left(\frac{1}{a}\right) \neq 0 \text{ dan } \frac{1}{\left(\frac{1}{a}\right)} = a$$

Pembuktian:

Membuktikan $\frac{1}{a} \neq 0$

Asumsikan $\left(\frac{1}{a}\right) = 0$

$$\Leftrightarrow 1 = a \left(\frac{1}{a}\right) \quad (M4)$$

$$\Leftrightarrow 1 = a (0) \quad (\text{asumsi } \frac{1}{a} = 0)$$

$$\Leftrightarrow 1 = 0 \quad (\text{Kontradiksi M3})$$

$\therefore$ Jika $\frac{1}{a} = 0$ maka $1 \neq 0$. Dengan demikian $\frac{1}{a} \neq 0$ terbukti.

Membuktikan $\frac{1}{\frac{1}{a}} = a$

Karena $\frac{1}{a} \neq 0$, dan $\left(\frac{1}{a}\right) \cdot a = 1$

$$\Leftrightarrow \left(\frac{1}{a}\right) \cdot a = 1 \quad (M4)$$

$$\Leftrightarrow a = \frac{1}{\frac{1}{a}} \quad (\text{kedua ruas dibagi } \frac{1}{a})$$

$\therefore$ Terbukti bahwa $a \neq 0 \Rightarrow \left(\frac{1}{a}\right) \neq 0$ dan $\frac{1}{\left(\frac{1}{a}\right)} = a$

$$(2) a \cdot b = a \cdot c \text{ dan } a \neq 0 \Rightarrow b = c$$

Pembuktian:

$$b = 1 \cdot b \quad (M3)$$

$$\Leftrightarrow b = \left(\frac{1}{a} \cdot a\right) \cdot b \quad (M4)$$

$$\Leftrightarrow b = \frac{1}{a} \cdot (a \cdot b) \text{ (M2)}$$

$$\Leftrightarrow b = \frac{1}{a} \cdot (a \cdot c)$$

$$\Leftrightarrow b = \left(\frac{1}{a} \cdot a\right) \cdot c \text{ (M2)}$$

$$\Leftrightarrow b = 1 \cdot c \text{ (M4)}$$

$$\Leftrightarrow b = c \text{ (M3)}$$

$\therefore$ Terbukti

(3) $a \cdot b = 0 \Rightarrow$ berlaku salah satu

$$a = 0 \text{ atau } b = 0$$

Pembuktian :

Asumsikan $a \neq 0$ dan dibuktikan bahwa $b = 0$

$$\frac{1}{a} (a \cdot b) = \left(\frac{1}{a} \cdot a\right) b \text{ (M2)}$$

$$\Leftrightarrow \frac{1}{a} (a \cdot b) = 1 \cdot b \text{ (M4)}$$

$$\Leftrightarrow \frac{1}{a} (a \cdot b) = b \text{ (M3)}$$

Karena $a \cdot b = 0$, maka :

$$\frac{1}{a} (a \cdot b) = \frac{1}{a} \cdot 0 = 0$$

Sehingga :

$$b = 0$$

$\therefore$ Terbukti

Asumsikan $b \neq 0$ dan dibuktikan bahwa $a = 0$

$$\frac{1}{b} (a \cdot b) = a \left(\frac{1}{b} \cdot b\right) \text{ (M2)}$$

$$\Leftrightarrow \frac{1}{b} (a \cdot b) = a \cdot 1 \text{ (M4)}$$

$$\Leftrightarrow \frac{1}{b} (a \cdot b) = a \text{ (M3)}$$

Karena $a \cdot b = 0$, maka :

$$\frac{1}{b} (a \cdot b) = \frac{1}{b} \cdot 0 = 0$$

Sehingga :

$$a = 0$$